

DAT 5564: Database Design and SQL

Fall 2025, Mini A

Lab #4: SQL Part 3

INSTRUCTIONS

1. This is a group assignment to be completed during the lab.
2. Please include only your Student IDs on the submission.
 - a. This submission will be submitted online on Canvas.
 - b. Each member of your group needs to submit, even though you are working in a group.
 - c. This submission is automatically graded.
 - d. The submission is due on Canvas.

SESSION OUTLINE

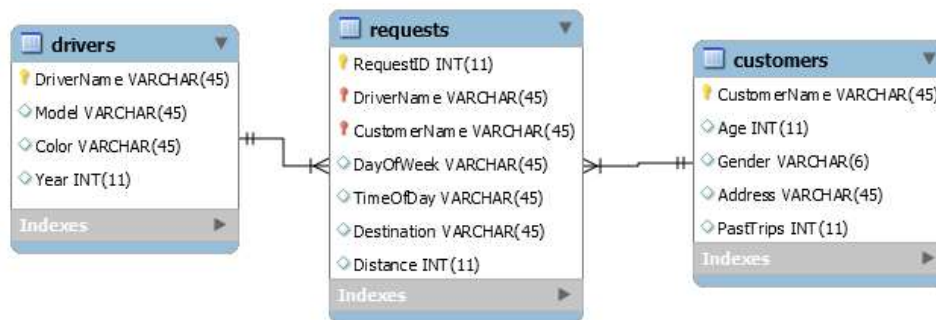
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| • Lecture: Review of SubQueries | 15 minutes |
| • Assignment Questions #1 – #14 | 40 minutes |
| • Solutions to some questions | 25 minutes |

ASSIGNMENT

Use the Lyft database to answer all questions in this assignment.

The database is posted on Canvas

The E/R diagram for this database is below:



- 1) Which drivers did not have any requests? Use “Having” in SQL to identify these drivers.
- 2) Calculate the number of requests for each day of the week. What is the least busy day?
- 3) Find the total number of unique customers who have driven in black cars. Separate these customers by gender. How many male customers were in black cars?
- 4) Calculate the average distance for requests in the “Evening” for each destination. Show only destinations with an average distance of more than 8 miles. Sort the results based on decreasing order of the average distance. Which destination has the shortest average distance?
- 5) Find the driver of the oldest car (Use a subquery). Your SQL code should show only the driver’s name.
- 6) Convert the previous query to a subquery in order to identify how many rides have been requested from that driver.
- 7) Are certain car colors more popular? For each car color, calculate: the number of rides, the number of cars, and the average number of rides per car. Include all cars in this query. Optional: calculate the average distance per car of that color.
- 8) For each customer, find the total number of rides they had, both in the past and in the current week. Is there any relationship between the number of rides a customer had in the past and the number of rides in the current week? Sort your results by the total number of rides and return only the 12 customers with the fewest rides, in total.
- 9) Convert the previous question (Question #8) into a subquery to find the average total number of trips for each address. Remember that the address is the city the customer lives in. Sort results in increasing order of average number of trips.
- 10) List the number of requests by time of day on each day. List only times with more than 5 requests.
- 11) What is the average age of travelers to each destination? Order the destinations from the oldest to the youngest.
- 12) List destinations that have more than 4 ride requests. Include the destination and the number of requests. Sort the results based on the decreasing order of the number of requests.
- 13) List the RequestID, CustomerName, Distance, and any two other attributes for trips that are longer than the average trip. Return only the 10 shortest trips.